QISKIT FALL FEST 2026 · IIIT-DELHI · TRACK 3 QUANTUM SIMULATION · INTERMEDIATE + ADVANCED ANALYSIS

Dynamics Under Noise

From an 8-spin Trotter study to a **certified 100-qubit kicked-Ising experiment** on a calibrated IBM Heron noise model

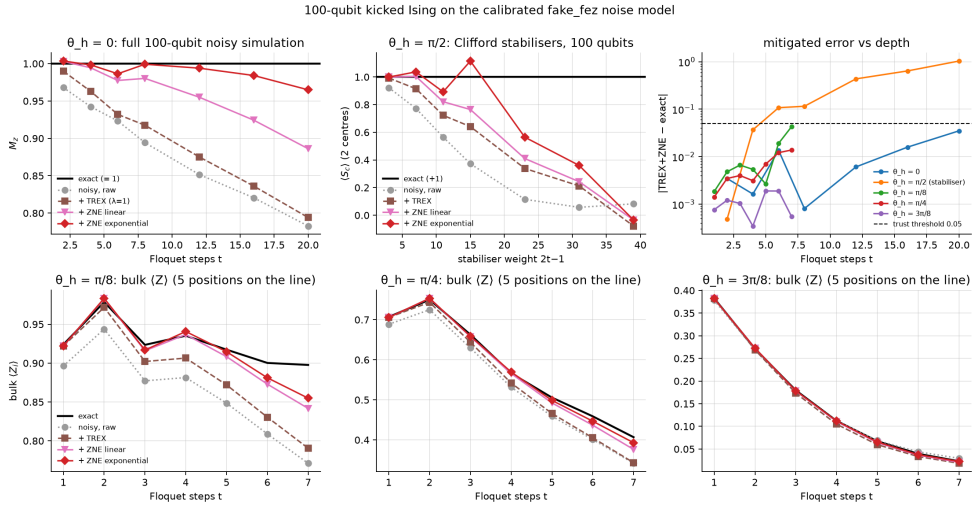
100 qubits
error-aware line on heavy-hex,
0 SWAPs

1881 CZ
deepest circuit, 2q depth 38, 20 Floquet steps

3 exact checks
free fermions + Pfaffians · light cone · Clifford, all at N = 100

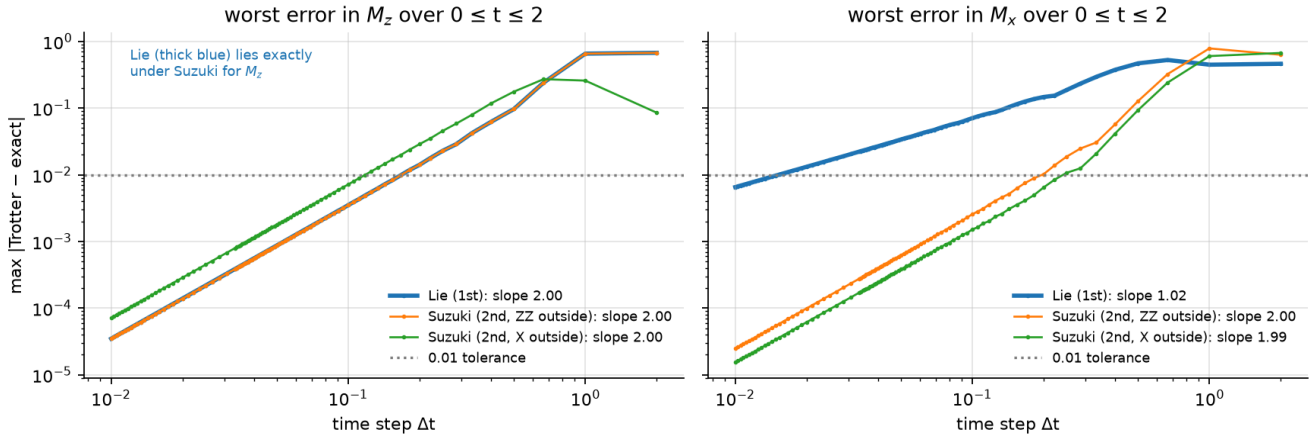
t = 20 recovered
θ_h=0 M_z within 0.05 of exact at 1881 CZ (100-qubit noisy emulation); stabilisers only to t = 4 (weight ≤ 7)

- **Beginner:** for Z observables from |0…0⟩, Lie-Trotter *is* Suzuki-Trotter, exactly. Mx needs 2nd order: 12× fewer CX for 0.01 accuracy (154 vs 1834).
- **Intermediate:** 2nd-order accuracy at 1st-order CZ cost. Our own TREX + ZNE on FakeFez noise. Under noise, finer Trotter steps stop helping.
- **Advanced:** 100-qubit kicked Ising (0 SWAPs) on a noise model from the chosen line's calibration, with TREX + ZNE, certified against exact answers at **every** depth. QPU jobs prepared and validated; no QPU access before the deadline.



BEGINNER · 8 SPINS · J = h = 1 · t ≤ 2

Why the "right" Trotter step depends on what you measure



formula	obs	largest Δt (err<0.01)	CX Qiskit	CX merged
lie	Mz	0.1667 (12 steps)	168	154
lie	Mx	0.0152 (132 steps)	1848	1834
suzuki	Mz	0.1667 (12 steps)	336	168
suzuki	Mx	0.1818 (11 steps)	308	154
strang	Mz	0.1176 (17 steps)	442	238
strang	Mx	0.2222 (9 steps)	234	126

- $|0\ldots0\rangle$ is an eigenstate of every Z_iZ_{i+1}, so the **first ZZ layer is a phase**.
- A **final ZZ layer commutes** with every Z-basis observable.
- Strip both boundary pieces: Lie and Suzuki apply the **same gates** for M_z, $\langle Z_0Z_r\rangle$ (difference 1e-16). Lie is secretly 2nd order (slope 2.00).
- For M_x the final ZZ(Δt/2) is the correction: Lie is 1st order (slope 1.02), Δt = 0.015 vs 0.18.
- Merging adjacent half-steps: 2nd order at **1st-order CX cost**.

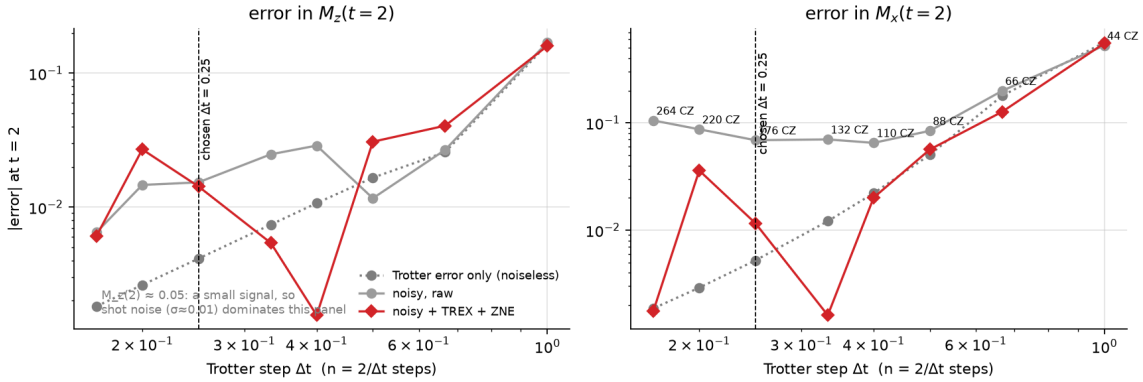
> Cheapest correct choice: Lie for Z-only; merged Suzuki (Δt = 2/11, 154 CX) for M_x; Δt = 2/12 (168 CX) for both.

INTERMEDIATE · CUT DEPTH BEFORE YOU MITIGATE

Hardware-aware compilation: every CZ removed is error you never mitigate

12-spin TFIM to t = 2 on FakeFez	CZ	2q depth	SWAPs	ESP
(a) Qiskit SuzukiTrotter, default layout	322	322	0	0.298
(b) merged brickwork, default layout	176	32	0	0.499
(c) merged brickwork, error-aware line	176	32	0	0.499
(d) merged brickwork, scattered qubits	747	184	190	0.000

- **Brickwork ordering**: even bonds, then odd. 2q depth per step is constant in N.
- **Merged half-steps** and a **dropped trivial first layer**.
- **Error-aware line**: beam search over the coupling map minimising Σ CZ + readout + SX infidelity from live calibration. Finds a 100-qubit path on 156-qubit Heron.
- Kicked Ising at $\theta_J = -\pi/2$: each RZZ is **one CZ**. 100 qubits, 20 steps: 1881 CZ, 0 SWAPs.



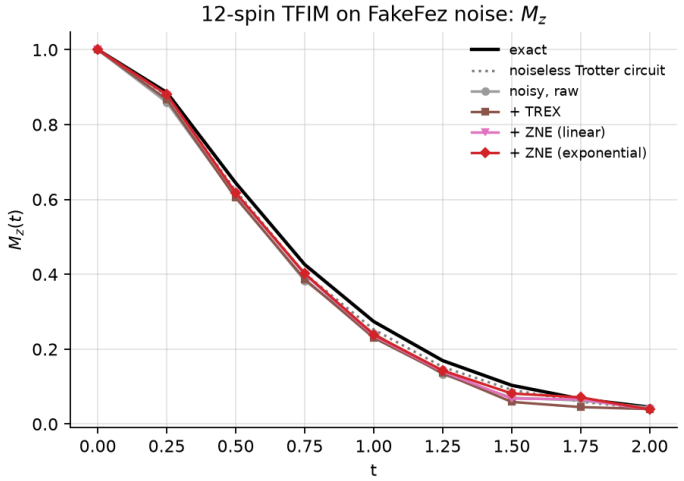
Noise-limited plateau: for Δt ≈ 0.25–0.4 the raw error is flat (≈0.07), and finer steps make it worse (0.105 at Δt = 0.167). After ZNE all Δt ≤ 0.4 agree within shot noise, so the Trotter-only Δt ≈ 0.15 buys nothing and costs 62% more CZ (286 vs 176).

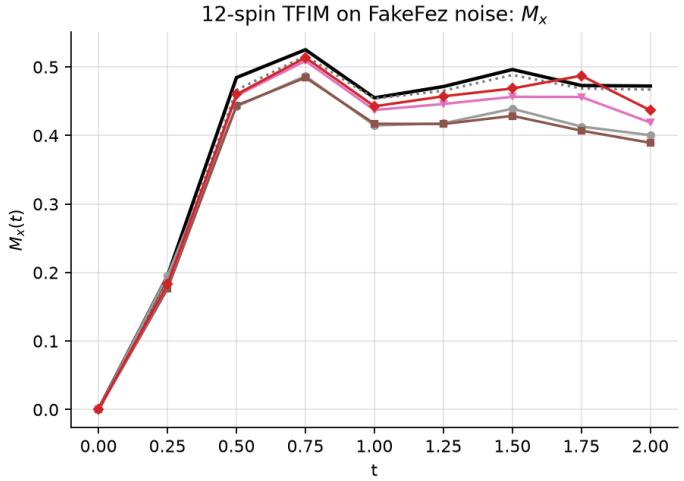
ESP = Π(1−ε_gate)·Π(1−ε_readout) from FakeFez calibration

§2.1, §2.4

INTERMEDIATE · 12 SPINS · FakeFez NOISE · OUR OWN MITIGATION CODE

Recovering the physics that noise hides





- **TREX**: random X before readout, undone classically, then rescaled by a twirled $|0\ldots0\rangle$ calibration.
- **ZNE**: every CZ folded to 3 or 5 CZs. Linear / exponential / Richardson to $\lambda \to 0$.
- **Twirling** is a no-op on Pauli noise (we check it). It matters on hardware, where errors are coherent.
- Compact noise model checked against the full FakeFez model.

mean |error| vs exact	M_z	M_x
raw	0.027	0.045
TREX	0.031	0.051
TREX + ZNE (linear)	0.021	0.026
TREX + ZNE (exp)	0.019	0.019

Job B (prepared, not run) would re-run these 8 circuits on a QPU for the noisy-sim vs hardware comparison

ADVANCED · HOW DO WE KNOW THE ANSWER AT 100 QUBITS?

Three independent exact certificates. No brute force.

method	what it gives	cost	valid
Free fermions + Pfaffians JW in X basis, a ghost spin turns Z_i into an even Majorana string; Wick → Pfaffian	every $\langle Z_i \rangle$, any $\langle Z_i Z_j \rangle$	$O(N^3)$ · 0.3 s at N = 100, t = 20	integrable 1D kicked TFIM, **any depth**
Light-cone window $\langle Z_k(t) \rangle$ depends only on [k−t, k+t]; one (2t+2)-qubit edge window + reflection gives all 100 sites	every $\langle Z_i \rangle$, $\langle Z_i Z_{i+1} \rangle$	2^{2t+2}, independent of N	any local circuit, $t \lesssim 11$
Clifford point $\theta_h = \pi/2$ $S = U Z_c U^\dagger$, weight 2t−1 Pauli	$\langle S \rangle = +1$ exactly	stabiliser tableau	any N, any depth

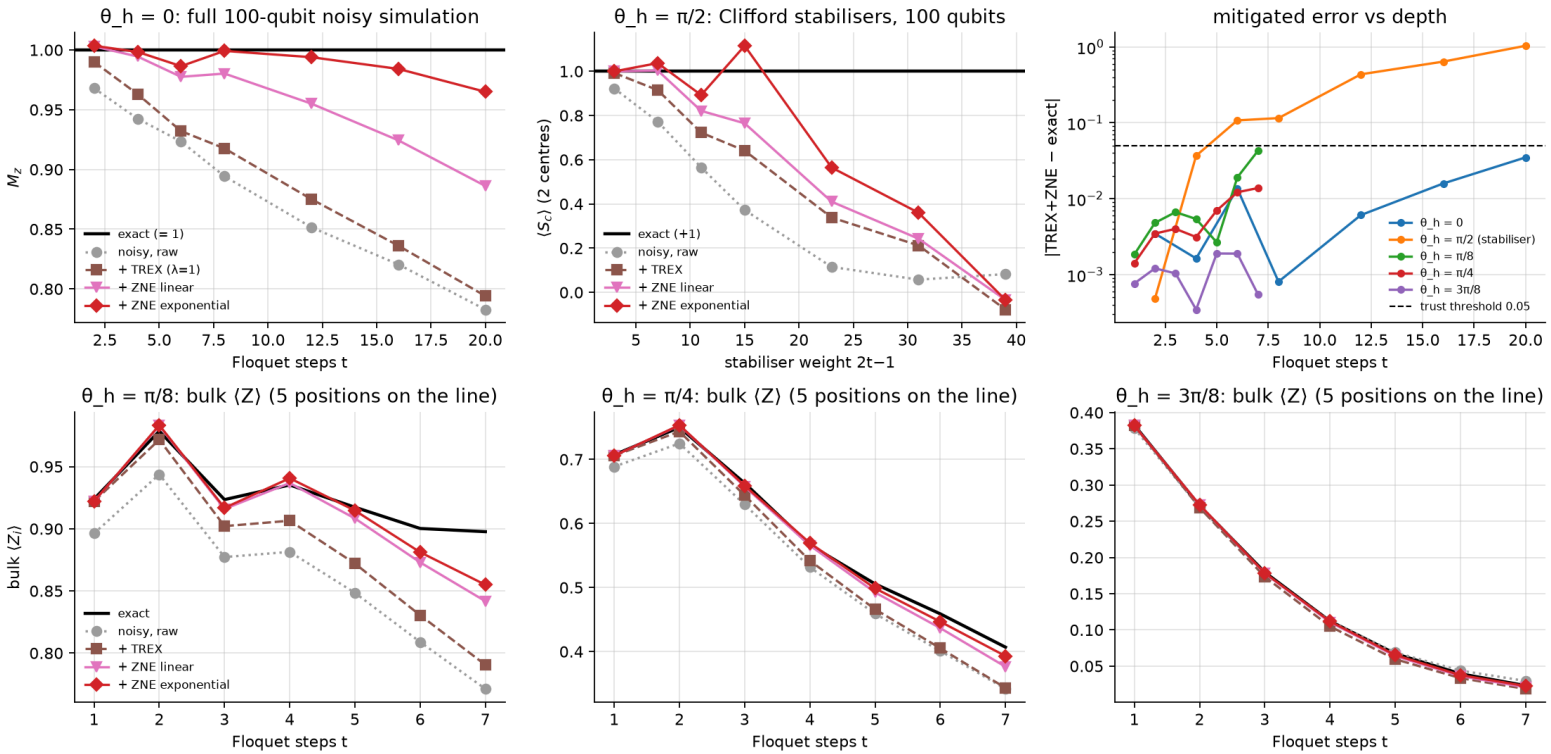
- Cross-validated: free-fermion vs light-cone at N = 100 agree to **~1e-14**, and both match statevector at N = 12.
- Kicked Ising (Kim *et al.*, Nature 2023) on a 1D chain: $U_F = \Pi\, R_X(\theta_h) \cdot \Pi\, R_{ZZ}(-\pi/2)$.
- 52 circuits: $\theta_h \in \{0, \pi/8, \pi/4, 3\pi/8\} \times t \in \{1 \ldots 8, 10, 12, 14, 16, 20\}$, plus 7 Clifford circuits.

> Honest framing: integrability is *why* we can certify 100 qubits at depth 20. We use it as a calibrated benchmark of the device + mitigation stack, not as a claim of quantum advantage.

ADVANCED · 100 QUBITS · CALIBRATED IBM HERON NOISE MODEL

The 100-qubit experiment, noisy, mitigated, and checked against exact answers

100-qubit kicked Ising on the calibrated fake_fez noise model

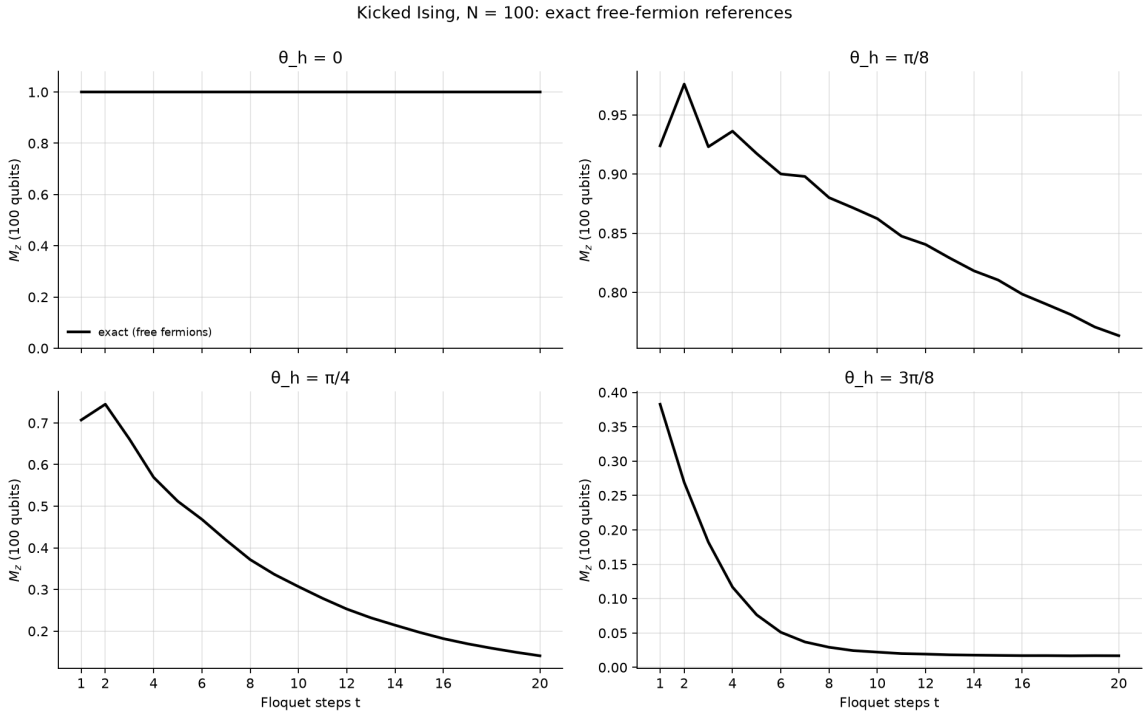


- **Clifford points** ($\theta_h = 0, \pi/2$): a stabiliser simulation of the **entire** noisy 100-qubit circuit is exact, so we run it at full size up to 20 steps / 1881 CZ, including folded ZNE circuits.
- **Other angles**: noise outside the light cone cannot matter, so we simulate (2t+1)-qubit windows at 5 positions on the line. They agree with full-chain density matrices to $\leq 4\times10^{-4}$.
- Noise: per-gate CZ/SX errors + asymmetric readout of the *chosen* 100 qubits (ibm_fez snapshot).

> θ_h=0, t=20 (1881 CZ): raw 0.78 → TREX+ZNE 0.96 (exact 1); weight-39 stabiliser: raw 0.08 → -0.03 (exact 1).

ADVANCED · TRUST HORIZON · QPU-READY

How far can we trust it, and how would we know without an exact answer?



- **Trust horizon** (|ZNE − exact| ≤ 0.05): 0: t=20, π/2 (stabiliser): t=4, π/8: t=7, π/4: t=7, 3π/8: t=7.
- **Which agreements count**: noise pulls ⟨Z⟩ towards 0, so $\theta_h = 3\pi/8$ (exact ≈ 0) "agrees" for free. High-signal observables ($\theta_h = 0$, π/8, Clifford stabilisers) are the real test.
- **Without an exact answer**: exponential vs linear ZNE spread, Clifford-point decay and light-cone windows (t ≲ 11).
- **QPU-ready**: 3 jobs (EstimatorV2 with TREX + twirling + XpXm DD + ZNE; SamplerV2 raw + readout calibration), PUBs validated, ≈ 4 Open Plan minutes (estimate); a ~1–2 min `--quick` subset. Every CZ count equals (t−1)·99, proving 0 SWAPs.

The horizon depends on observable weight. Local ⟨Z⟩ at 1881 CZ is recovered to 0.96 (exact 1), but a Pauli spread over 15+ qubits decays below the shot-noise floor at λ = 5 and ZNE cannot bring it back: weight-7 stabilisers pass, weight-11 and beyond fail.

exact $M_z(t)$ for the 4 angles on the left; free-fermion Pfaffians, 0.3 s each

LIMITATIONS · NEXT STEPS · REPRODUCIBILITY

Where it works, where it breaks, what we would do next

- **Classically easy by design**: integrability lets us certify. A longitudinal field h_z breaks it, leaving only light cone + Clifford + consistency checks.
- **Gate-folding ZNE** multiplies depth by 3–5. At $t \geq 16$ (1500+ CZ) $\lambda = 5$ nears the noise floor and the fits disagree.
- **No real-QPU data**: the noise model has no coherent errors, crosstalk or drift. Real hardware would decay faster. Twirling and DD target exactly those effects.
- **Next**: PEA-ZNE with learned layer noise (only 2 distinct CZ layers), fractional RZZ for the TFIM (halves depth), a non-integrable chain.

prepared QPU job	primitive	content
A	EstimatorV2: TREX, twirl, DD, ZNE	52 kicked-Ising + 7 Clifford circuits, 100 qubits
B	EstimatorV2: same options	8 TFIM circuits, 12 qubits
C	SamplerV2 (raw)	26 circuits + readout calibration

- Everything in one notebook that runs top to bottom (~5 min). QPU and long sweeps are cached in `results/`.
- `hardware/run_hardware.py`: dry-run / submit / fetch with the exact same circuits.
- Tutorial credit: BasQ Qiskit Fall Fest 2026 (B. Tirado). AI assistance: Claude, for coding and writing, with every number verified in-notebook.